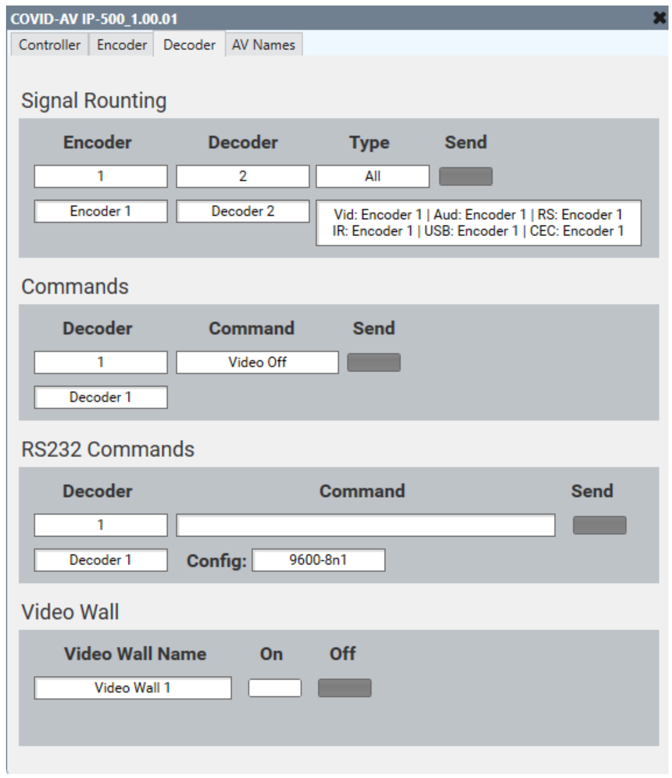
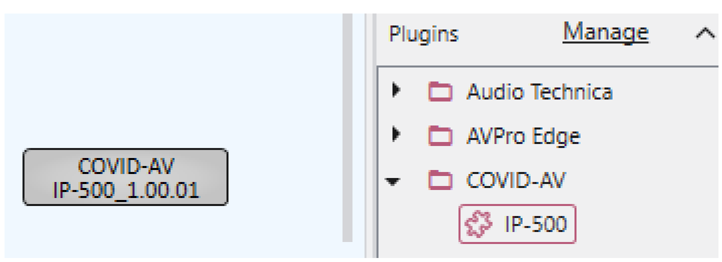
COVID-AV IP-500 Q-SYS Plugin
Version 1.00.01

The COVD-AV IP-500 Q-SYS Plugin makes the operation of the IP-500 system very easy when building the Q-SYS User Interface design. Programming can be done using the control pins for simple interfacing or for advanced users by using Component Controls.



To install the COVD-AV IP-500 Q-SYS Plugin, just doble-click on the Plugin or place it in your documents folder under QSC\Q-Sys Designer\Plugins. When you open Q-SYS Designer you will see it in Plugins.



Properties

This Plugin dynamically configures its inputs and outputs to the size you set in the properties.

First, you need to set up the properties of the Plugin. The properties items include:

- Connection - TCP or Serail
- IP – The IP Address
- PresetSize – The number of presets you are using.
- EncoderSize - The number of Encoders you are using.
- DecoderSize - The number of Decoders you are using.

Properties

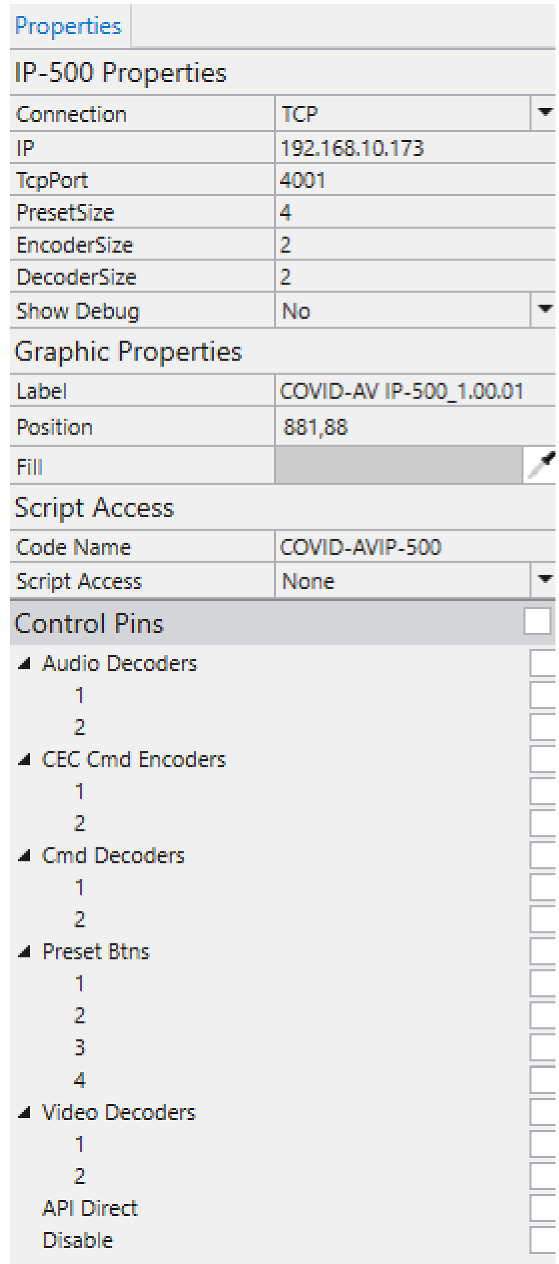
IP-500 Properties	
Connection	TCP
IP	192.168.10.173
TcpPort	4001
PresetSize	4
EncoderSize	2
DecoderSize	2
Show Debug	No
Graphic Properties	
Label	COVID-AV IP-500_1.00.01
Position	881,88
Fill	
Script Access	
Code Name	COVID-AVIP-500
Script Access	None

Control Pins

For easy configuration of your User Control Interface, the Plugin uses Control Pins. These Control Pins are dynamically set by the size you set in properties.

The following Control Pins are used:

- Audio Decoders
    - These pins are used to route audio from an Encoder to a Decoder using the String value of the Encoder.
- CEC Cmd Encoders
    - These Pins are used to send defined CEC receiver commands as a string.
- CMD Decoders
    - These Pins are used to send defined Decoder commands as a string.
- Preset Btns
    - Select a preset as a Boolean button command.
- Video Decoders
    - These pins are used to route video from an Encoder to a Decoder using the String value of the Encoder.
- API Direct
    - Use this pin to send a direct API command. Do not include the “>” or the Delimiter.



Operation

The IP-500 Plugin will emulate the control of the system when it is not connected. This allows for testing of the User Control Interface. When connected the Network Connect Status LED will be green.

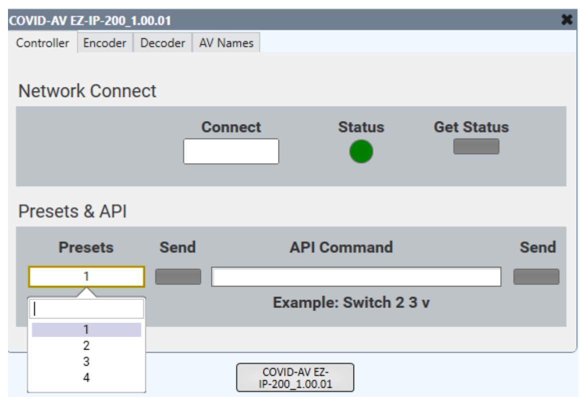
There are four pages associated with the Plugin:

1. Controller
2. Encoder
3. Decoder
4. Status

Controller Page

Network Connect

Although the IP-500 Plugin automatically connects at startup, the Connect button will allow you to Disconnect and Connect to the IP-500 system. A Green Status LED indicates that the Plugin is connected to the system. Once connected, the Plugin requests the Status from the IP-500 system. This status includes the routing information and the names of the Encoder and Decoders. If needed, the status can be requested by pressing the Get Status button.



Preset & API

One of the very powerful features of the IP-500 system is the Preset controls. Within the system up to 100 presets can be created to route AV and control displays.

In some situations, you may want to send a direct API command to the IP-500 system. The manual lists all the commands. Just enter the command without the “>” and termination.

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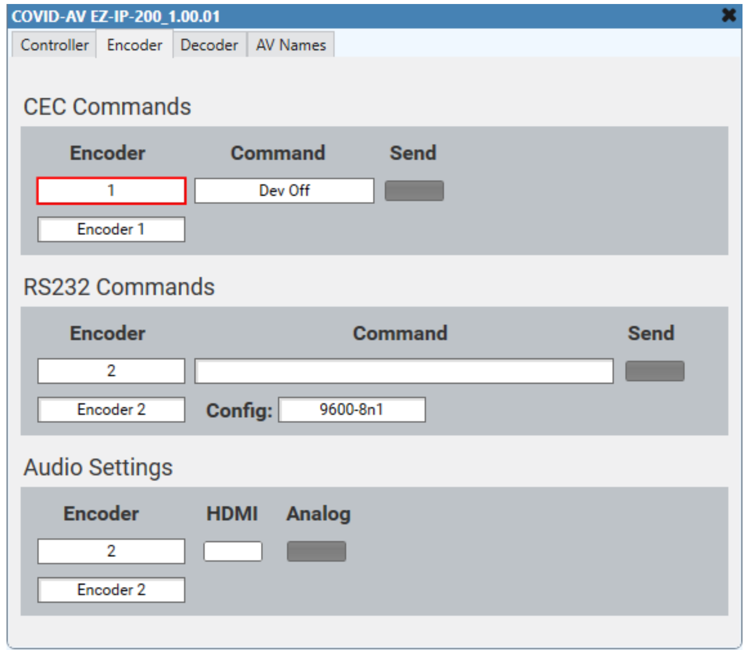
Encoder Page

This page allows the following:

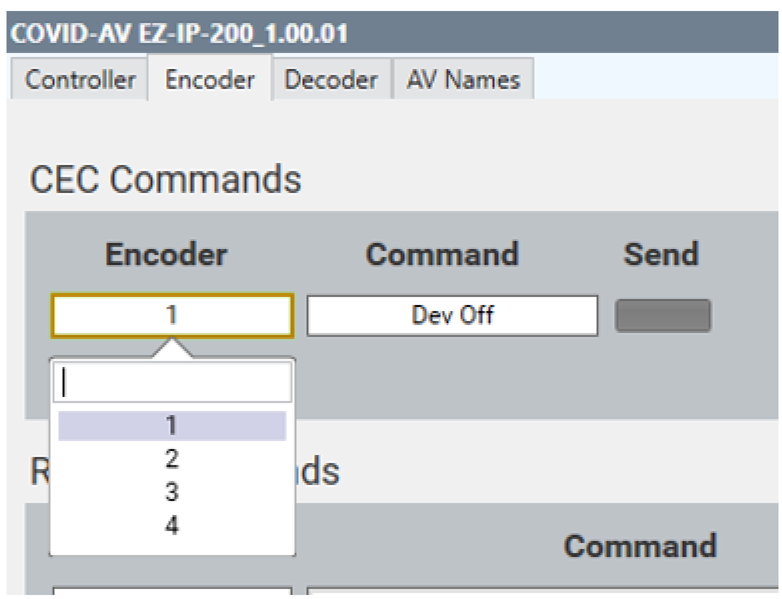
- Sending selected CEC receiver commands to an Encoder
- Direct RS232 commands to an Encoder
- Setting the audio input of an Encoder

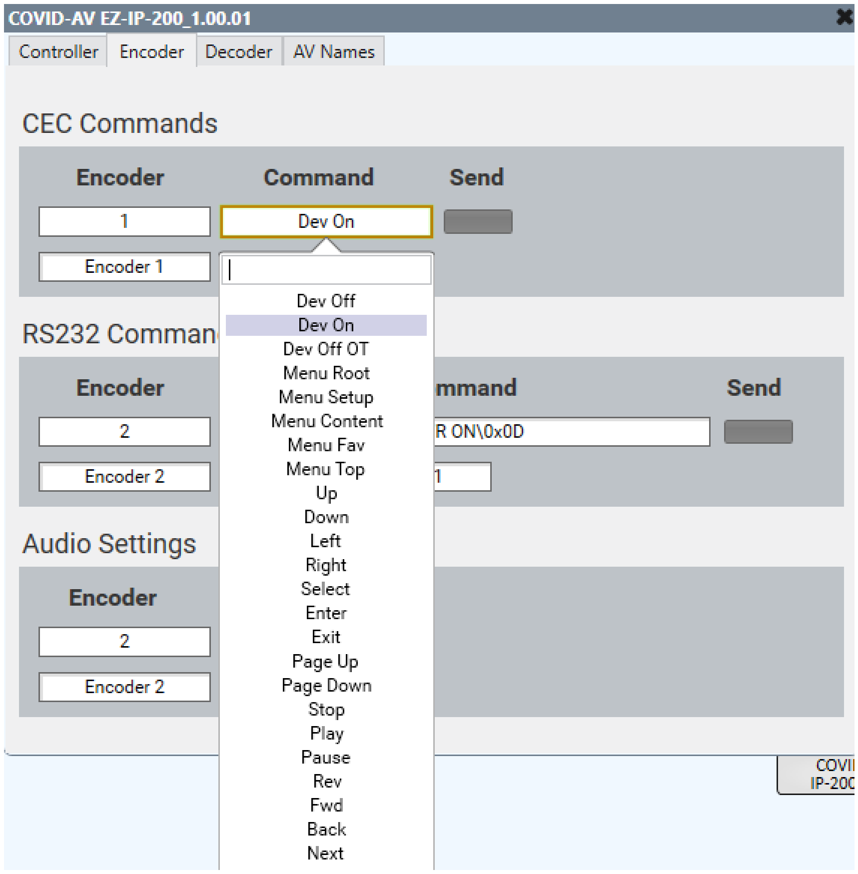
The name under the Encoder selection displays the name of the Encoder in the IP-500 system.



CEC Commands

This page allows sending selected CEC receiver commands to an encoder to control a Blu-ray and ATV device. First select the Encoder followed by the command.





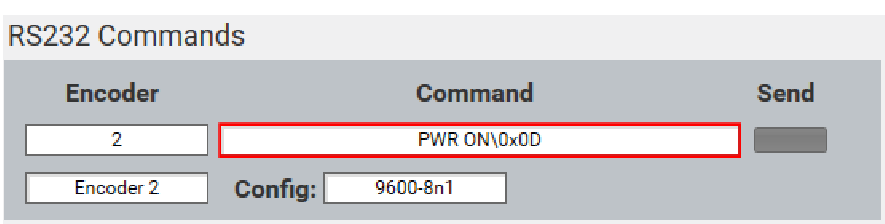
Encoder CEC Commands

RS232 Commands

RS232 Commands can be sent directly to the RS232 port on the Encoder. For UCI operation it is best to use the API Direct Control Pin. Example send to Encoder ID 2 “Power On\x0D\x0A”: “RS232E 2 9600-8n1 Power On\x0D\x0A”. Direct commands do not provide feedback.

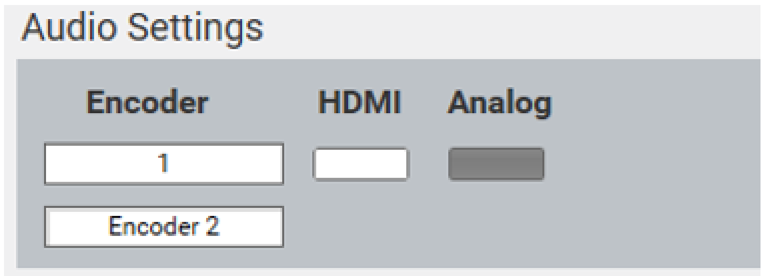
To send a command:

- Select the Encoder
- Enter the baud, data bits, parity and stop bits
- Enter the device command using String commands and “\0x##” for hex numbers
- Press Send



Audio Settings

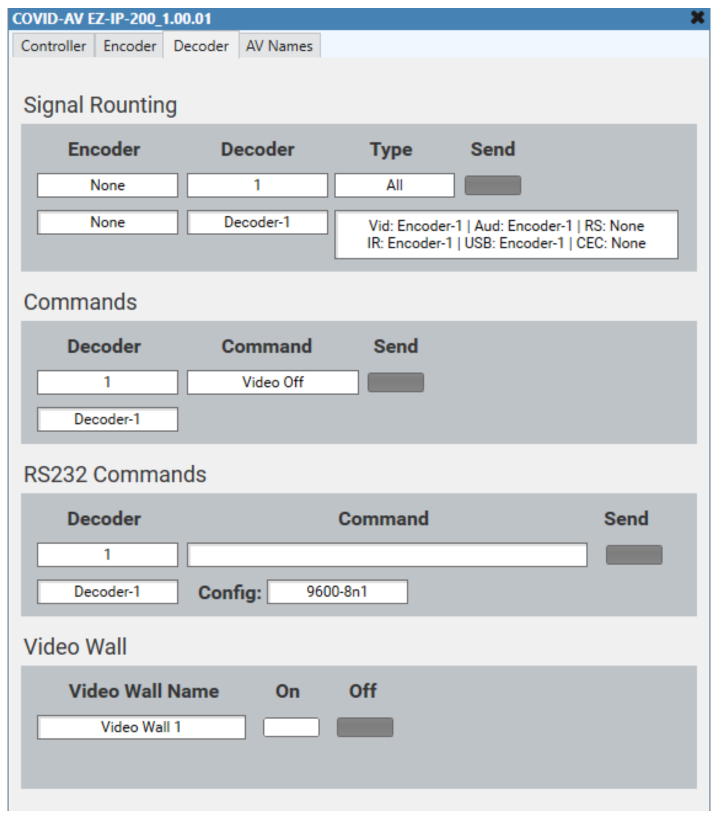
The audio settings allow for selecting the audio type for the Encoder. The selection can be standard HDMI or Embed Analog audio.



Decoder Page

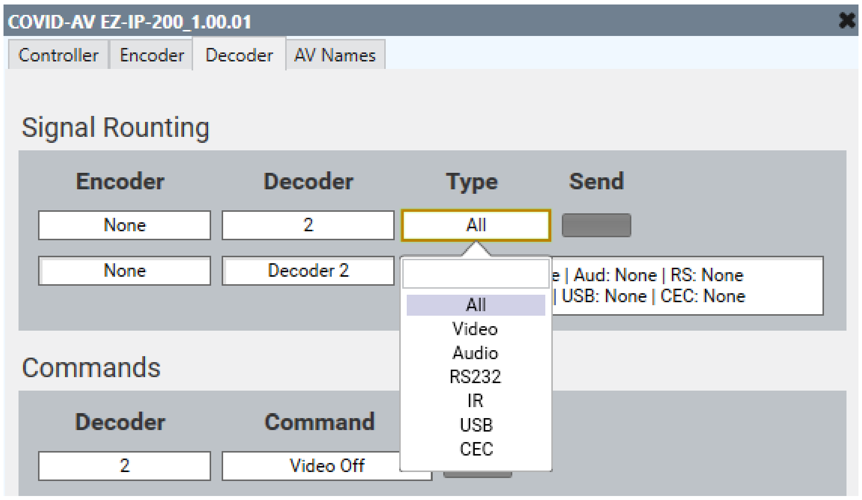
The Decoder Page allows for the following:

- Signal Routing
- Commands
- RS232 Commands
- Video Wall



Signal Routing

Routing is very simple by just selecting the Encoder ID followed by where you want it to go by the Decoder ID, the type of routing and then the Send button. A status box shows the current route for the Decoder. If the Encoder is set to None, the Route will be cleared in the Decoder.

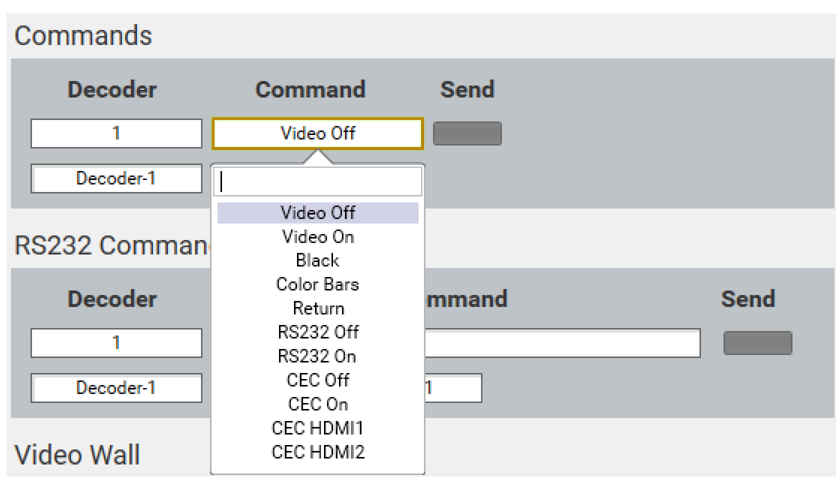


Commands

A list of commands can be selected from to send to a selected Decoder.

Commands:

- Video Off and On – Used to disable and enable the video output of the Decoder. This can be used to power on and off monitors that do not have control but can power by signal detection.
- Black, Color Bar and Return – These can be used for testing. Black can be used for picture blanking. To return to video, select Return.
- RS232 Off and On – Select these to send the RS232 power commands set in the Decoder.
- CEC Off and On – Select these to send the CEC power commands set in the Decoder.
- CEC HDMI 1&2 – These commands send the CEC input command of a display.

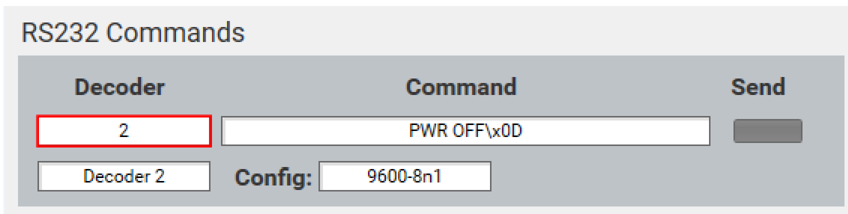


RS232 Commands

RS232 Commands can be sent directly to the RS232 port on the Encoder. For UCI operation it is best to use the API Direct Control Pin. Example send to Decoder ID 10 “Power On\x0D\x0A”: “RS232D 10 9600-8n1 Power On\x0D\x0A”. Direct commands do not provide feedback.

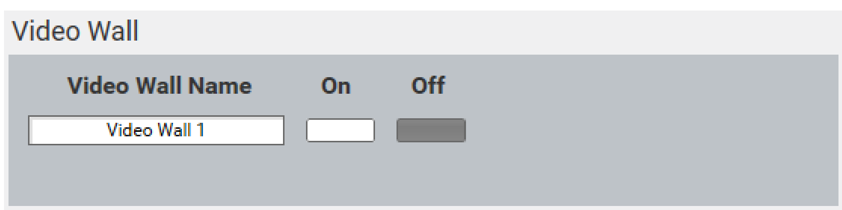
To send a command:

- Select the Encoder
- Enter the baud, data bits, parity and stop bits
- Enter the device command using String commands and “\0x##” for hex numbers
- Press Send



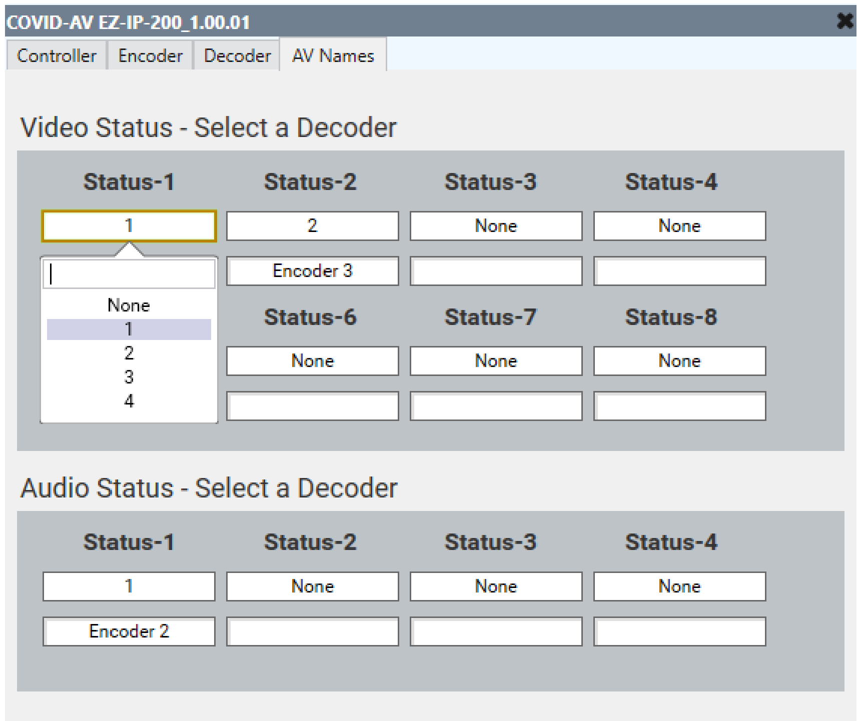
Video Wall

The operation of the Video Wall can be turned on or off. When on, the Video Wall is operational, and the Decoders are scaled to create a single image. When off, the Video Wall is disabled, and the Decoders are set to Single View allowing multiple displays like a quad view. Use the name assigned in the IP-500 system.



AV Names Page

This page allows you to monitor the routing of a Decoder. Select None for no status information. These can be used in the UCI to monitor what is sent to an output.



Video Status

There are up to 8 Decoders that can be monitored for their video routing.

Audio Status

There are up to 4 Decoders that can be monitored for their audio routing.

Control Pins

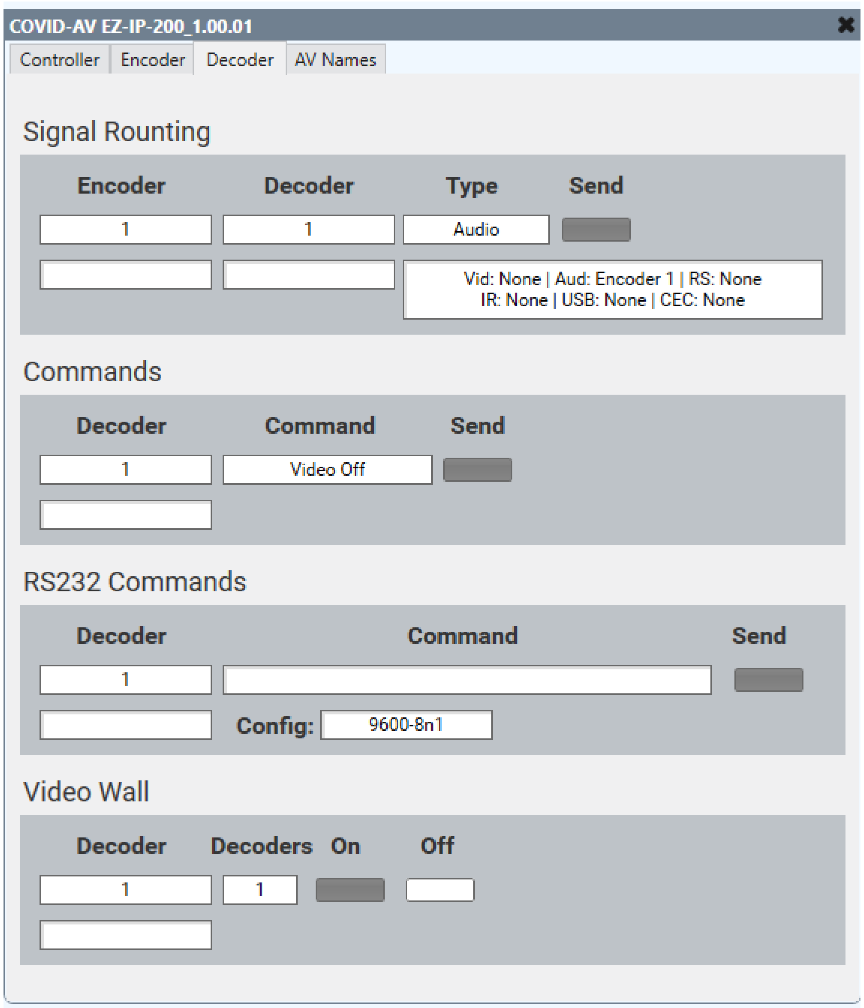
Control Pins are used to make it easy to interface the Plugin with the UCI.

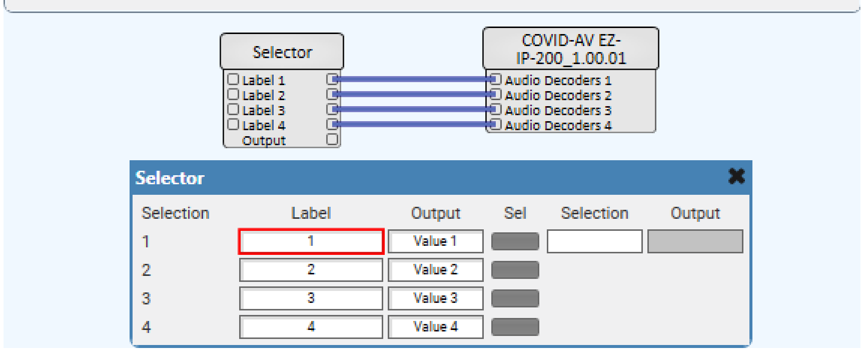
Audio Decoders

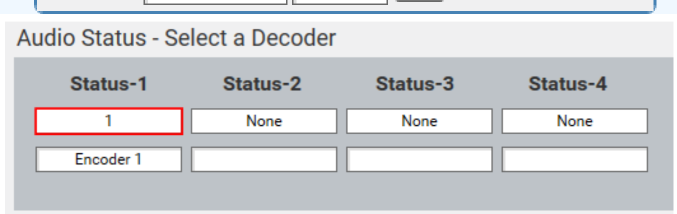
These pins are used to route audio from an Encoder to a Decoder using the String value of the Encoder.

Sending Encoder 1 Audio to Decoder 1.

Using labels on a Selector connected to the Audio Decoders you can enter the Encoder ID, and this will route that Encoder to the Audio Decoder. Entering “0” clears the Audio Decoder. In Signal Routing Aud. Encoder 1 is shown, and Audio Status-1 set to Decoder 1, Encoder 1 is shown on the AV Name Page.





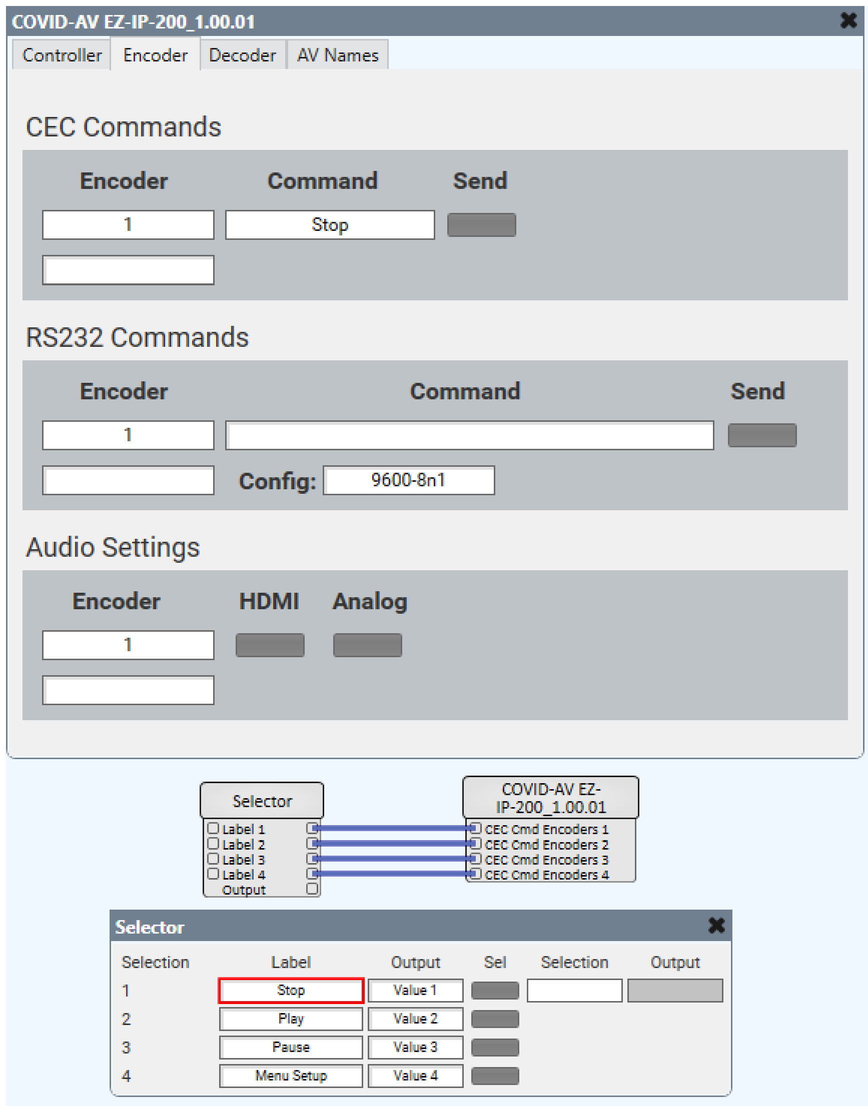


CEC Cmd Encoders

These Pins are used to send defined CEC receiver commands as a string.

List of CEC Commands:

- Dev Off
- Dev Off
- Dev Off OT
- Menu Root
- Menu Setup
- Menu Content
- Menu Fav
- Menu Top
- Up
- Down
- Left
- Right
- Select
- Enter
- Exit
- Page Up
- Page Down
- Stop
- Play
- Pause
- Rev
- Fwd
- Back
- Next

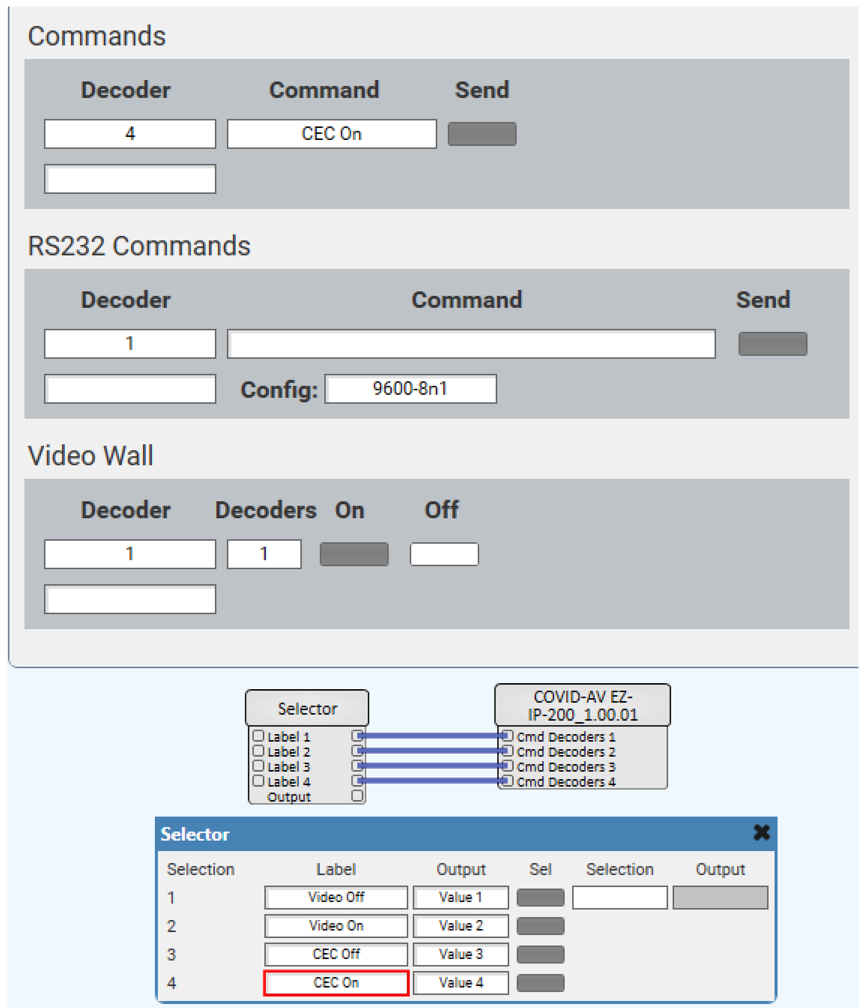


CMD Decoders

These Pins are used to send defined Decoder commands as a string.

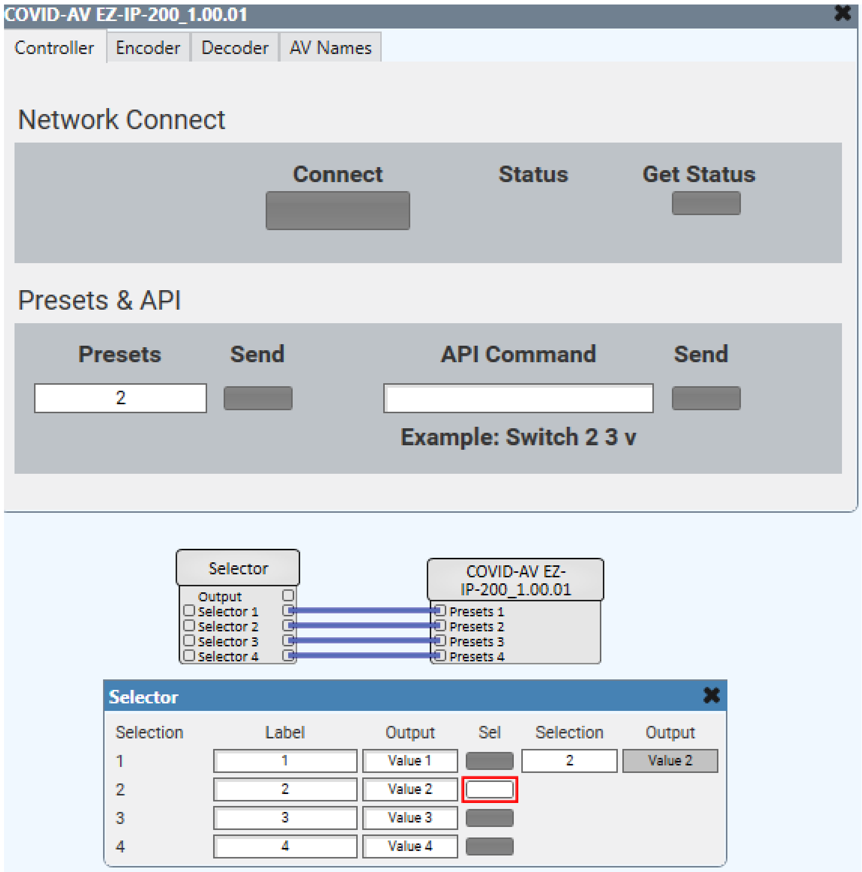
List of Decoder Commands:

- Video Off
- Video On
- Black
- Color Bars
- Return
- RS232 Off
- RS232 On
- CEC Off
- CEC On
- CEC HDMI1
- CEC HDMI2



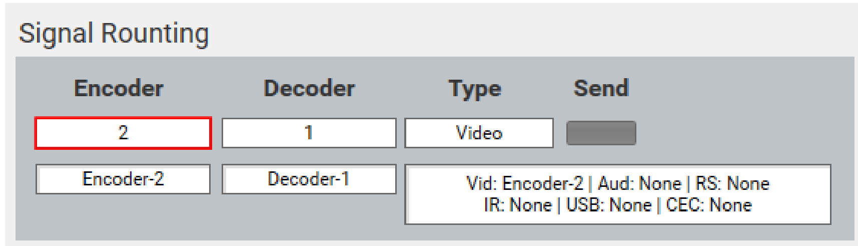
Preset Btns

Select a preset as a Boolean button command, such as a selector.



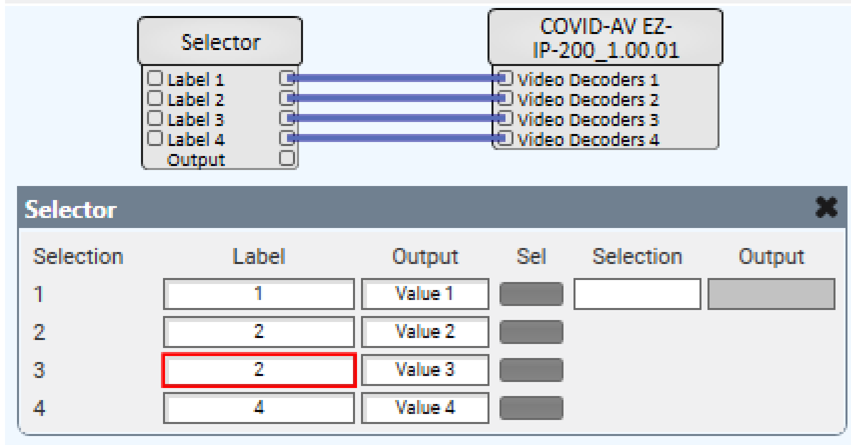
Video Decoders

These pins are used to route video from an Encoder to a Decoder using the String value of the Encoder.

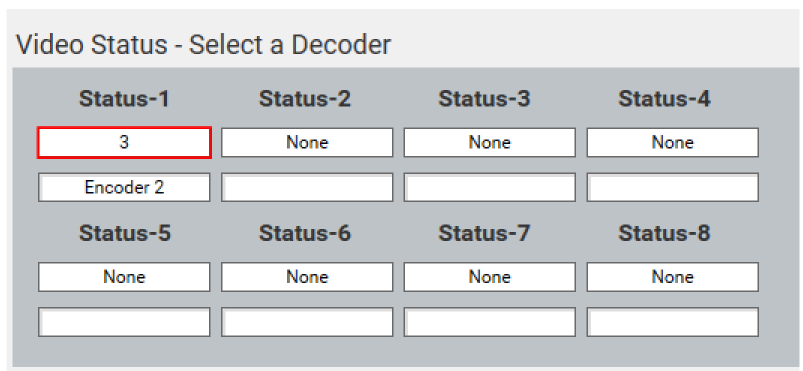


Using labels on a Selector connected to the Video Decoders you can enter the Encoder ID, and this will route that Encoder to the Video Decoder. Entering "0" or "None" clears the Video Decoder.

Sending Encoder 2 Video to Decoder 3.



In Signal Routing Vid. Encoder 2 is shown, and Video Status-1 set to Decoder 3, Encoder 2 is shown on the AV Name Page.



API Direct

Use this pin to send a direct API command. Do not include the “>” or the Delimiter.

Command sent: “RS232D 1 9600-8n1 Power on\x0D\x0A”

All commands can be found the COVID AV EZ-200 manual.